Centrifugal Confinement for Fusion

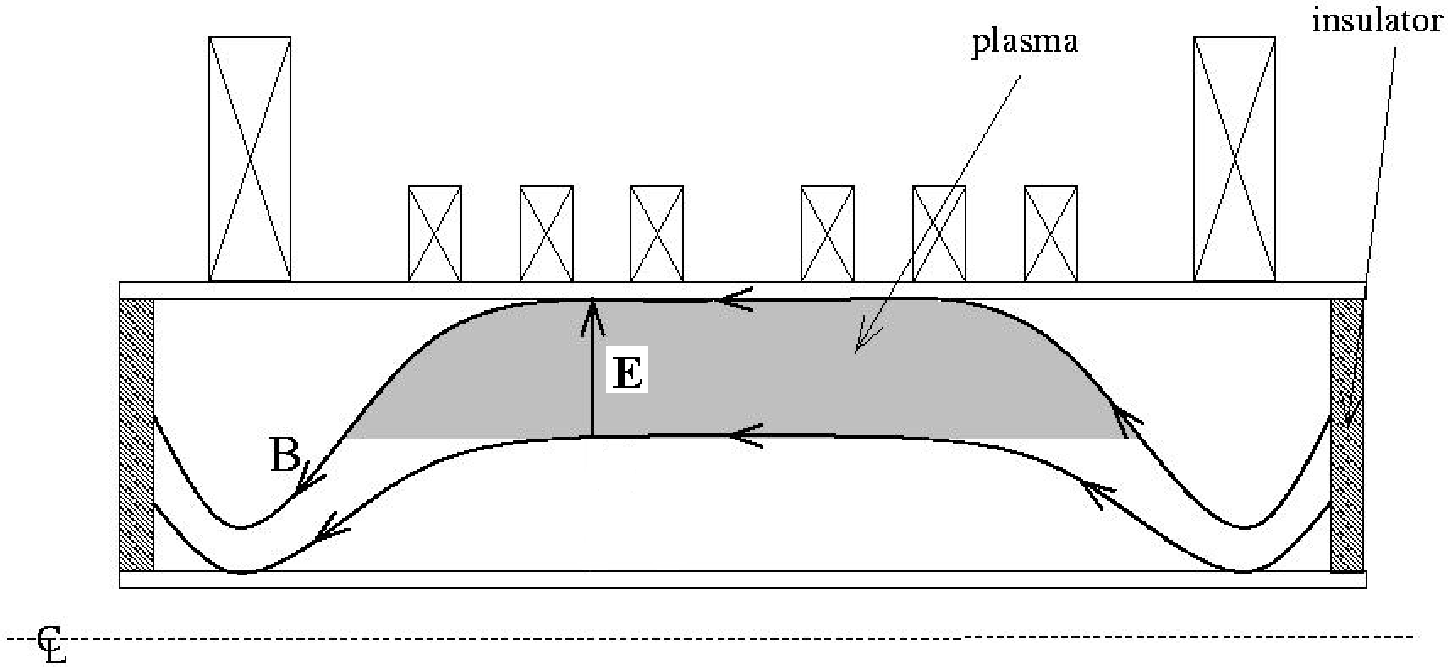
Adil Hassam and Rick Ellis

University of Maryland, College Park

Presentation to IEC US-Japan Workshop 2018

From Madison Workshop 2017

Basic Idea



- centrifugal forces => axial confinement
- rotation shear => stability to interchanges

Hassam, AB, Comments Plasma Phys Cont. Fus., 18, 263, 1997
Ellis, RF; Hassam, AB; Messer, S; et al. PHYSICS OF PLASMAS 8, 2057, 2001

Next few slides show why need $M_s > 1$

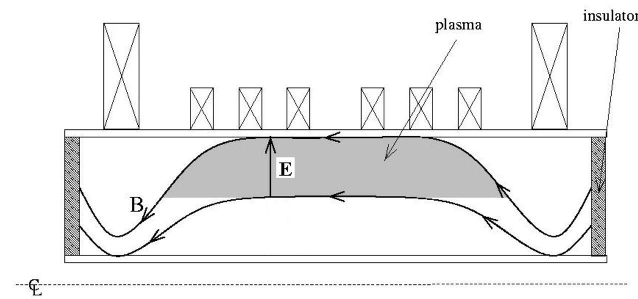
- **Sonic Mach number, M_s, is the Figure of Merit**

for Equilibrium, Stability, and Lawson Breakeven.

$$M_s > 1$$

MHD Centrifugal Confinement
=> need high Mach number

$$\mathbf{B}.\nabla\, p = -\,\mathbf{B}.[nm\, \mathbf{u}.\nabla\, \mathbf{u}]$$



$$\mathbf{p} \quad : \quad \mathbf{nmu^2}$$

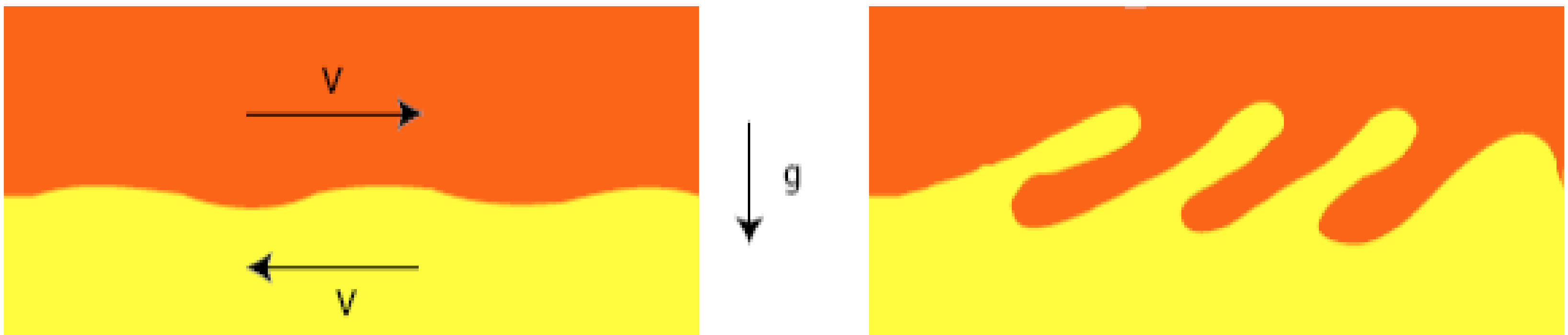
$$\mathbf{1} \quad : \quad \mathbf{u^2/c_s^2 = M_s^2}$$

"gravitational" scale height ~ $\mathbf{1/M_s^2}$

=> $M_s > 1$

V'to stabilize interchanges

=> M_s > 1



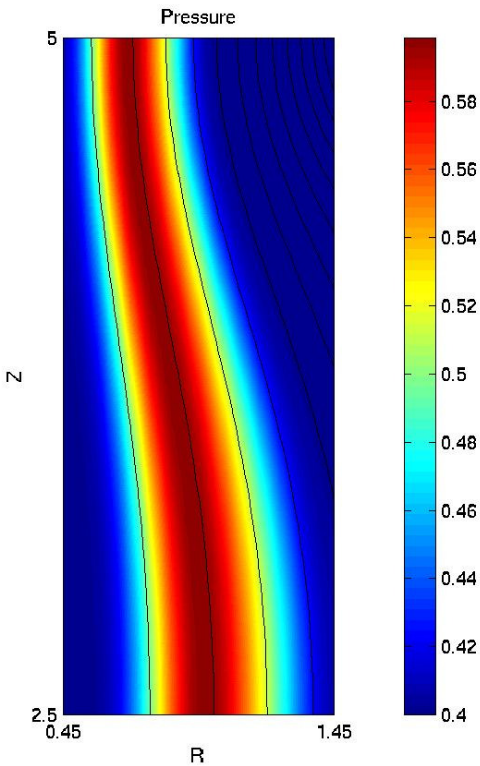
$$V' > \gamma_{int} \, [ln \, R_\mu]^{1/2}$$

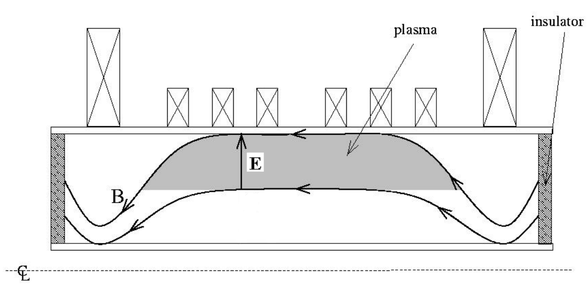
Hassam, Phys Fluids B, 4, 485 1992

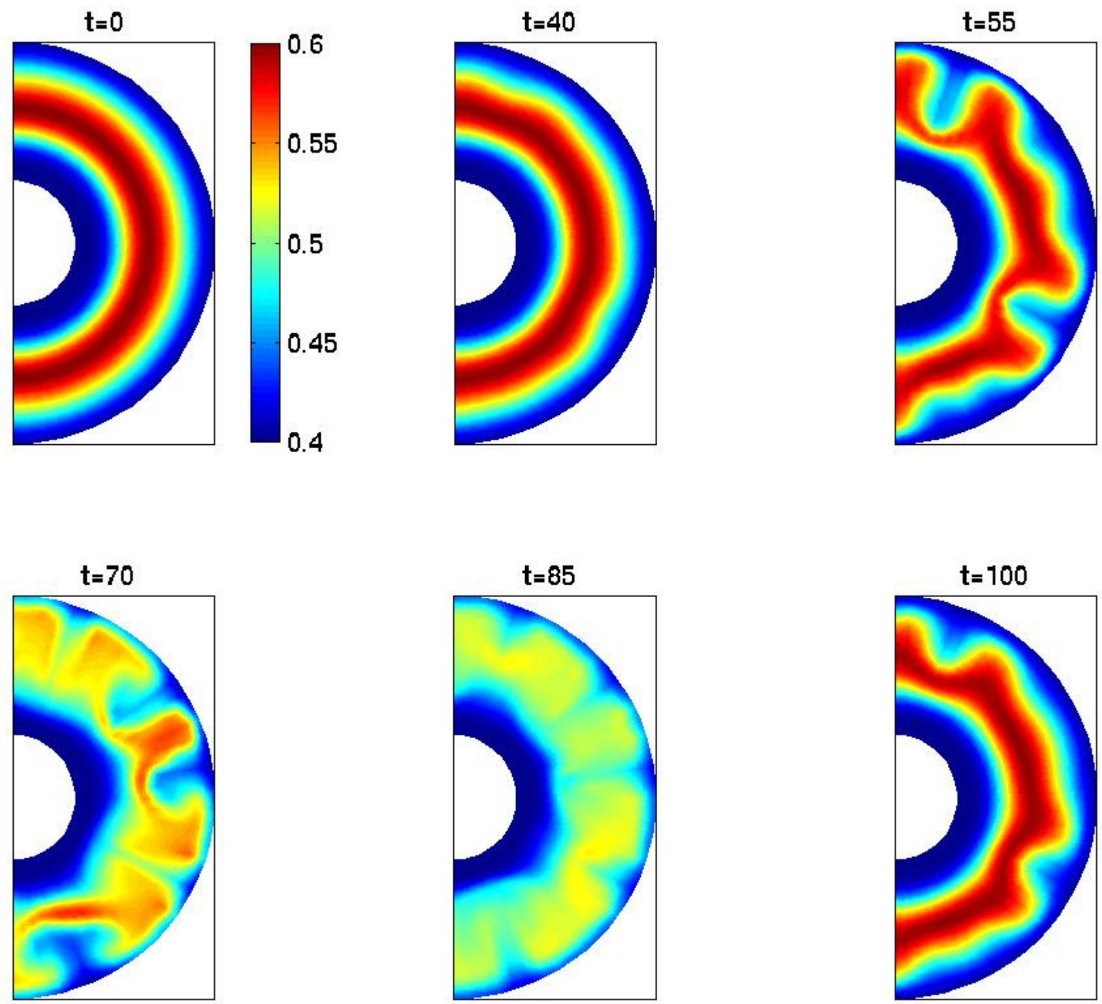
Interchanges are sonic;

smooth profiles => $M_s > 1$

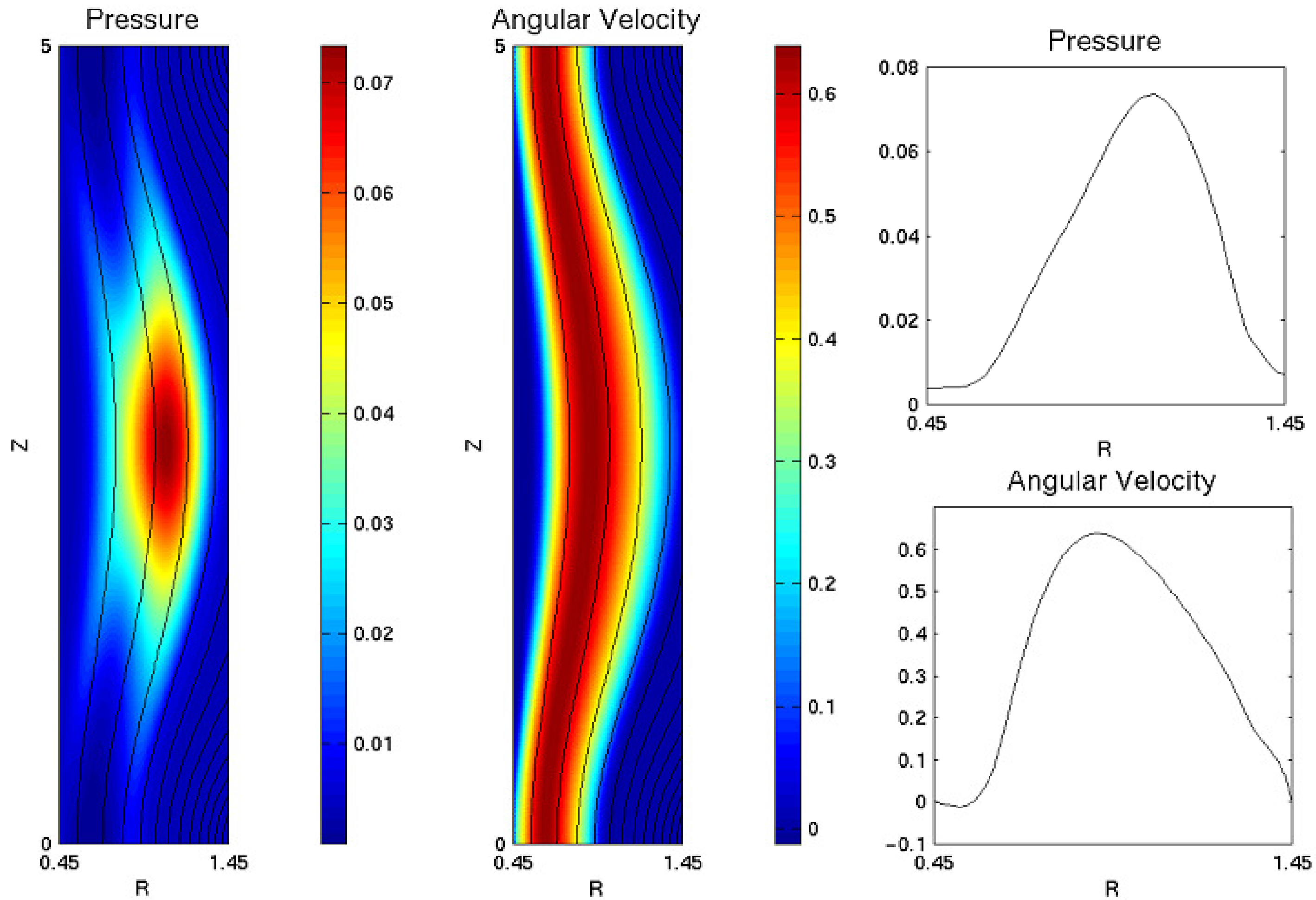
Simulation: Simple mirrors are unstable to flute interchanges





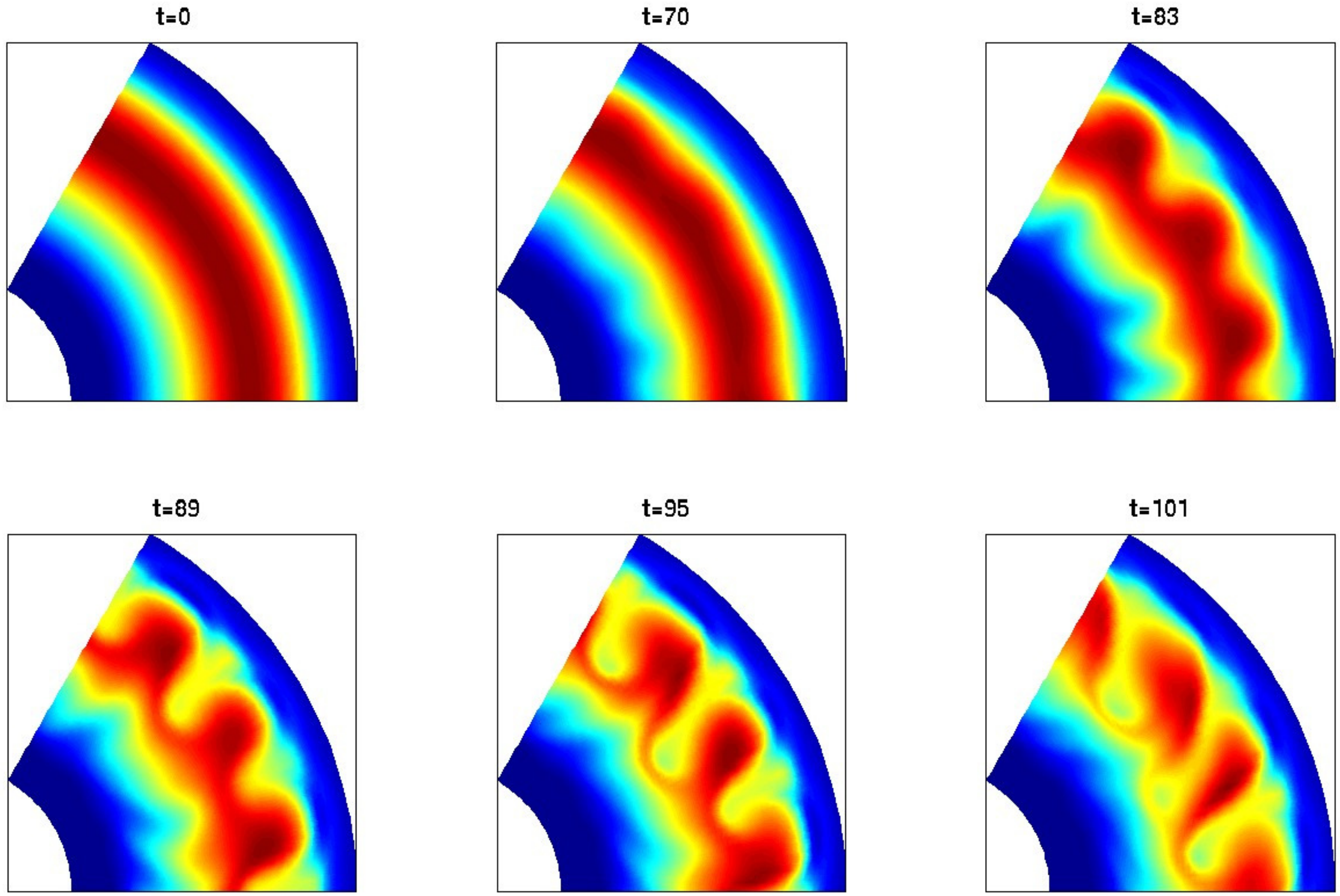


Supersonic rotation
=> centrifugal confinement



V’ shear has stabilized interchanges

- flutes appear if V’ artificially suppressed



Huang,Y-M, Hassam AB, Phys Rev Lett 87, 235002, 2001
Huang,Y-M, Hassam AB, Phys Rev Lett 87, 235002, 2001

TRANSPORT

- Cross field, classical?

- Along **B:**
  - Ions centrifugally confined
  - energetic electrons transfer heat
  - deep potential well, $e\varphi/T \sim M_s^2$
  - large Pastukhov factor

$$\tau_{||e} \sim \nu_{ee}^{-1} \; [M_s^2/4] \; \exp[M_s^2/4]$$

$M_s > 6$ => Lawson Condition

T. M. Antonsen, private communication
A. B. Hassam, Comments Plasma Phys. Control. Fusion 18, 275, 1997

The Maryland Centrifugal Experiment (MCX)

(2000 – 2010)

R. F. Ellis, A. B. Hassam

A. Case, D. Gupta, Y. Huang, J. Rodgers, C. Romero-Talamas, C. Teodorescu, A. DeSilva, R. Elton, H. Griem, P. Guzdar, R. Clary, S. Choi, R. Lunsford, A.S. Messer, R. Reid, G. Swan, I. Uzun-Kaymak, W. C. Young

University of Maryland, College Park

MCX Objectives

#1 Supersonic Rotation?

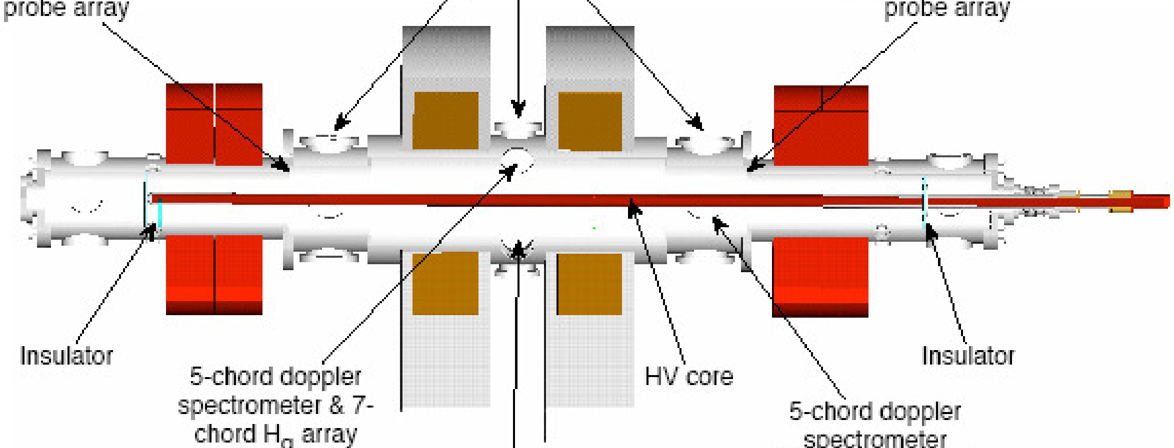
#2 MHD Stable?

#3 Centrifugally confined?

MCX

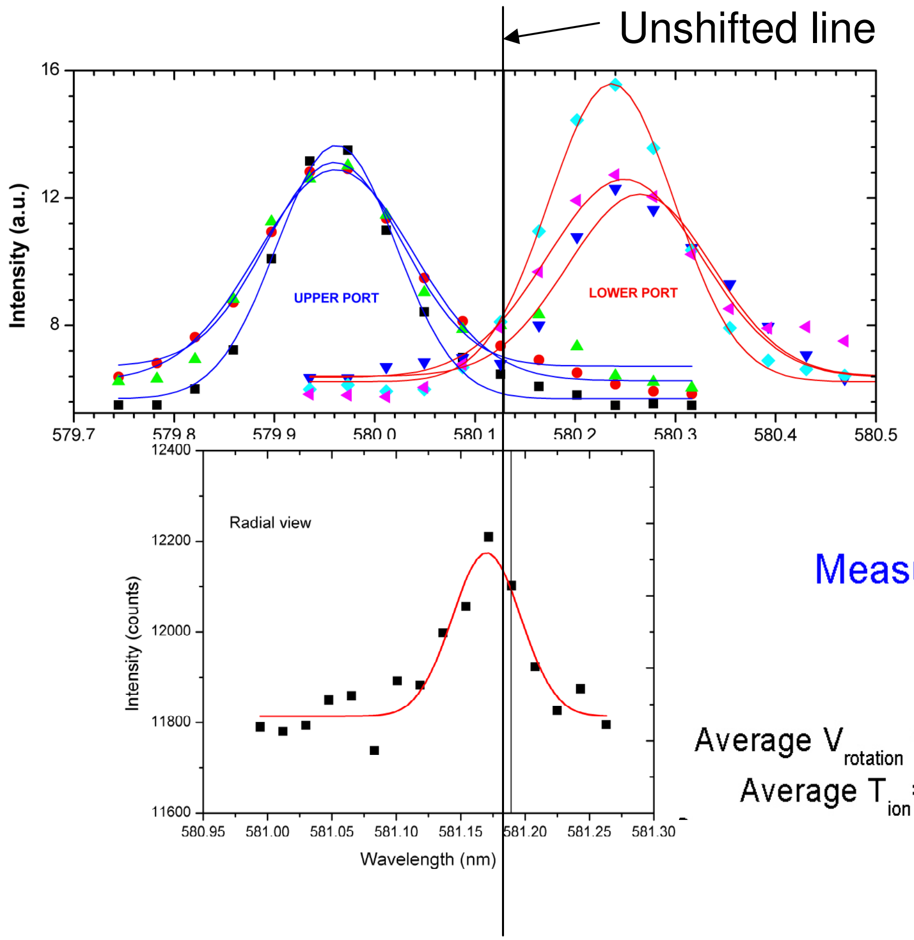
R_m ~ 9 (B_{mid} = 0.2 - 0.3 T)
Hydrogen, P_0 = 5-10 mtorr

n_i ~ few $\times 10^{20}$ m-3 (fully ionized)
T_i ~ 20 – 50 eV
V_{Bank} **5-20 kV**, pulse **1-10 ms**
$v_{rot} \equiv u_{ExB}$ ~ **100 km/s**



Messer, Case, Ellis, Gupta, Hassam, Lunsford, Ghosh, Elton, Griem, APS 2003

Goal#1: Doppler shifts show supersonic ExB rotation, in red and blue shifts (up and down)



CIV line at 5801.33

Measured at 550 μsec into the discharge

Integration time = 100 μsec

Average $V_{rotation}$ = 89 $\pm$ 5 km/sec
Average T_{ion} = 40 eV $\pm$ 10 eV

Average $V_{rotation}$ = 60 $\pm$ 5 km/sec
Average T_{ion} = 40 eV $\pm$ 10 eV

$M_s \sim 1 - 3$

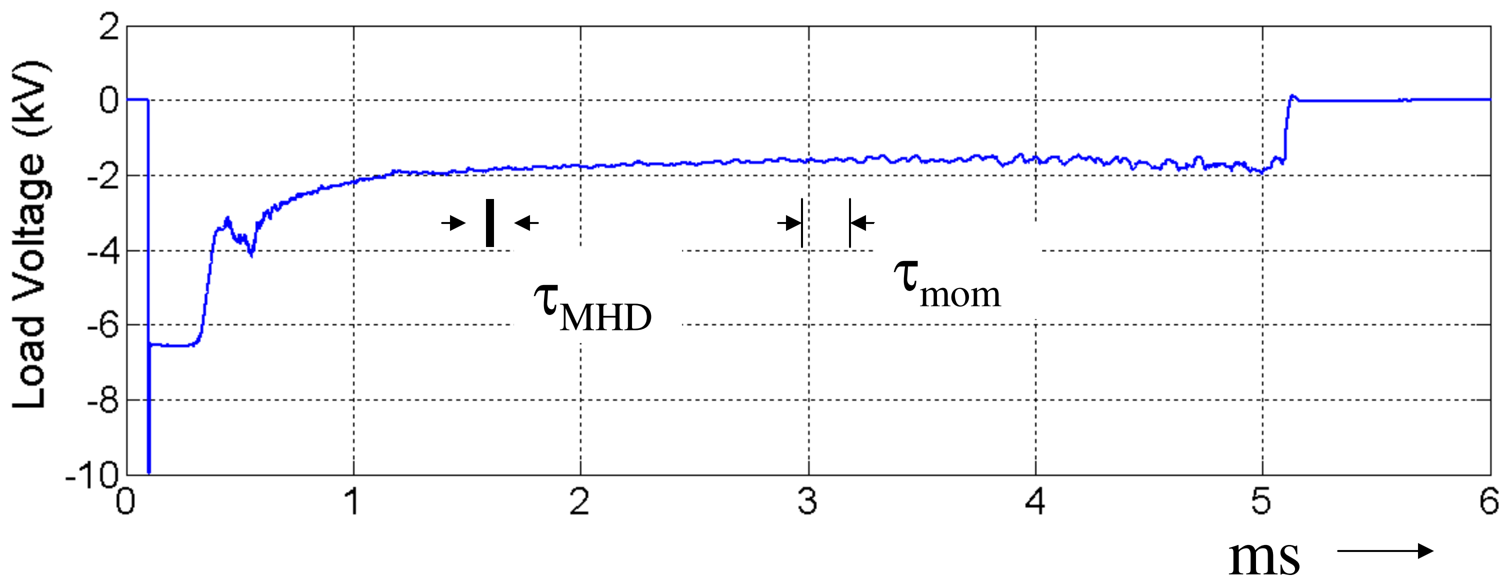
Ghosh, J; Elton, RC; Griem, HR; et al. Phys Plasmas 13, 022503, 2006

Goal #2:

MHD Stable?

Indirect Yes

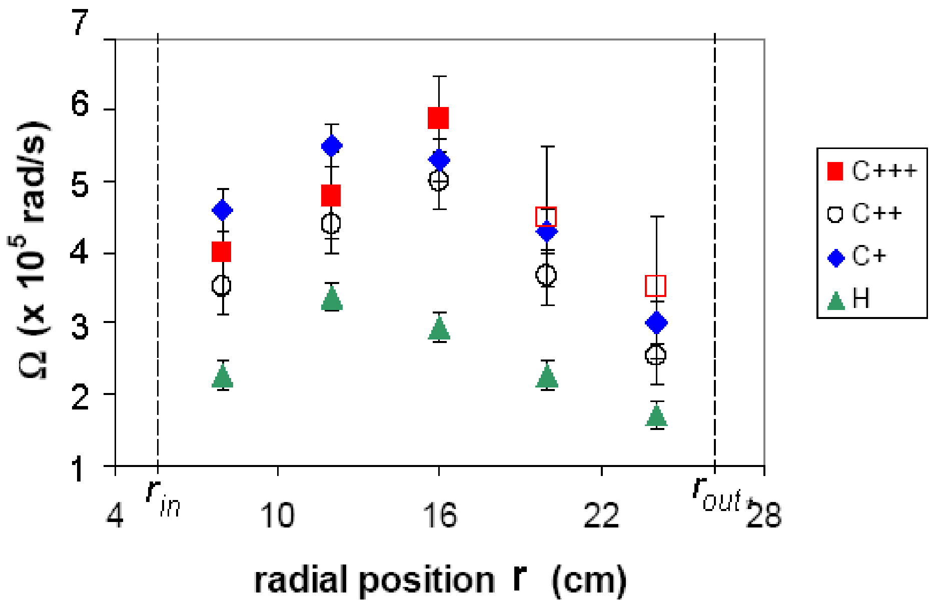
Voltage across plasma remains steady for 1000's of MHD instability times



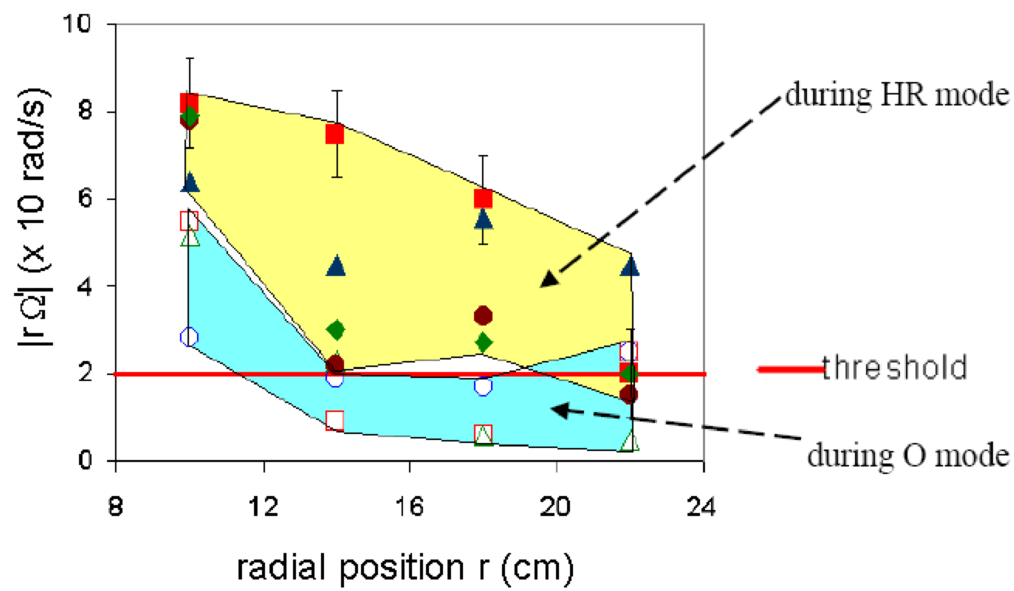
- MHD instability growth time τ_{MHD} ~ 2 - 20μs
- Measured momentum confinement time τ_{mom} ~ 200-800μs
- No "major disruptions" => **MHD Stable?**
- m=2 wobble ~ 1 cm

V’ shear is large enough to stabilize interchange modes

- Flow profiles independent of charge, consistent with **E**×**B** drift

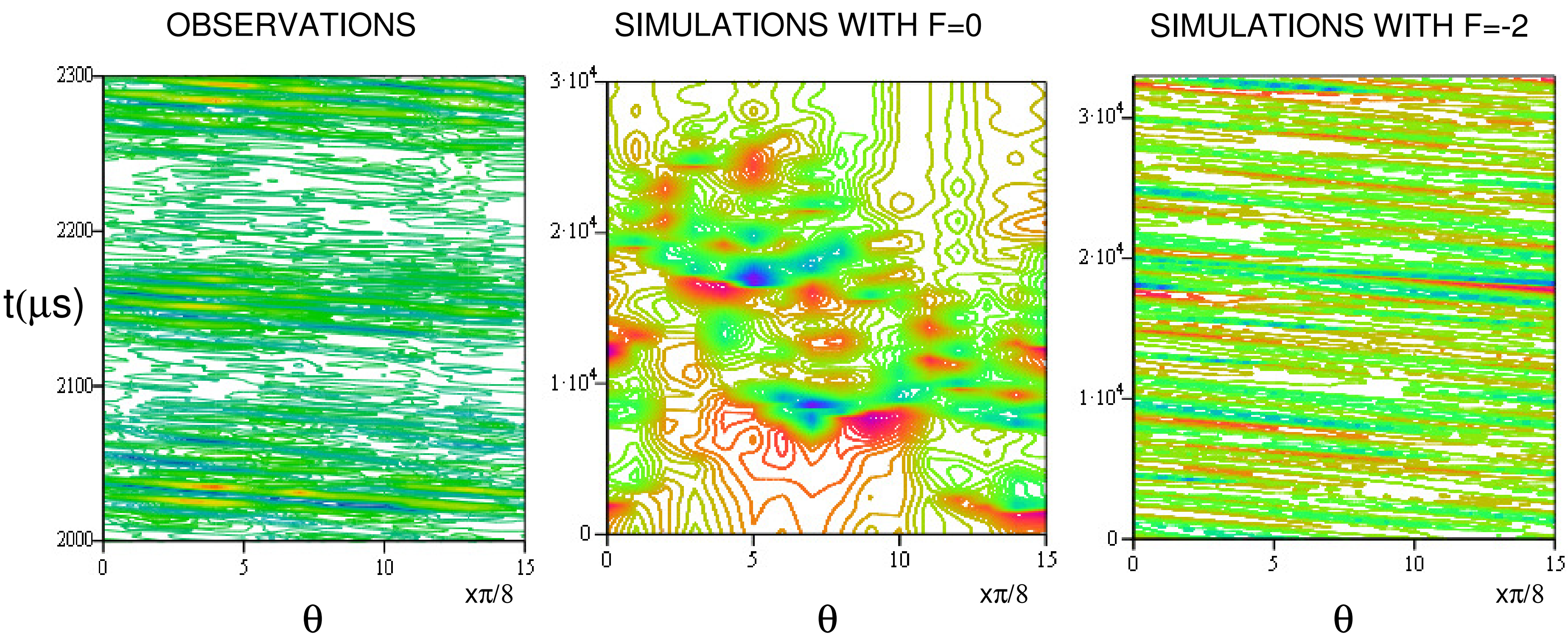


- Stability threshold exceeded



From C. Teodorescu, 2006 ICC Workshop, Austin, TX

Magnetic fluctuations: observations and simulation



- Simulation without imposed azimuthal flow (F=0) shows "bloby" structures with no clear direction of propagation
- Simulations with flow (F=-2) shows propagation features similar to observations.

Uzun-Kaymak, et al, Physics of Plasmas 15, 112308 (2008)

Goal #3:

Centrifugally confined?

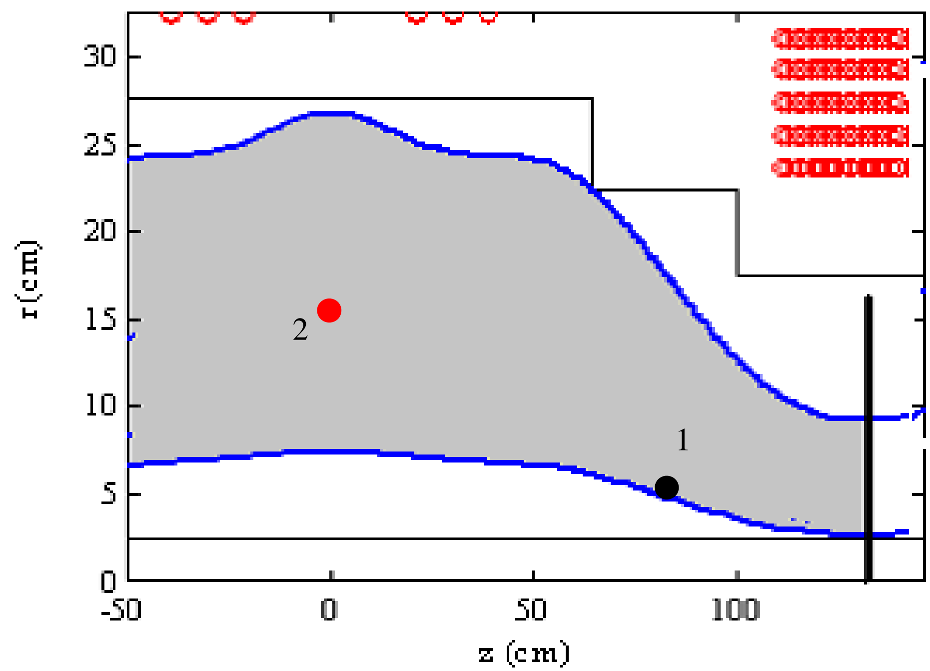
Yes

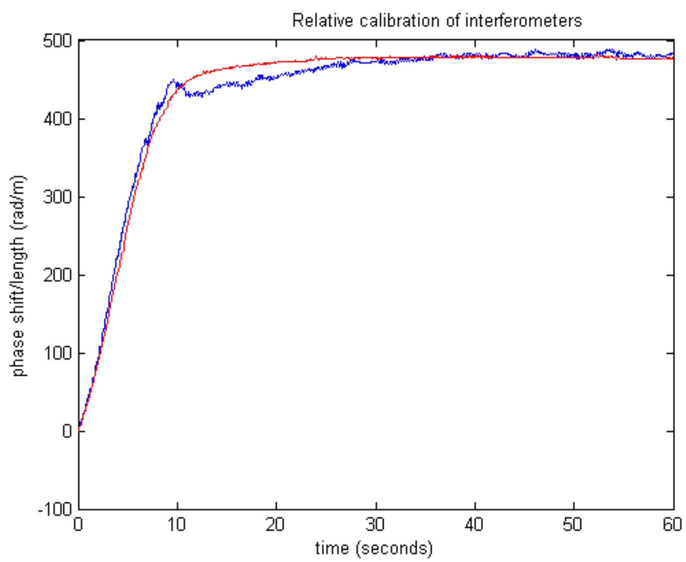
Midplane and axially off-midplane interferometers

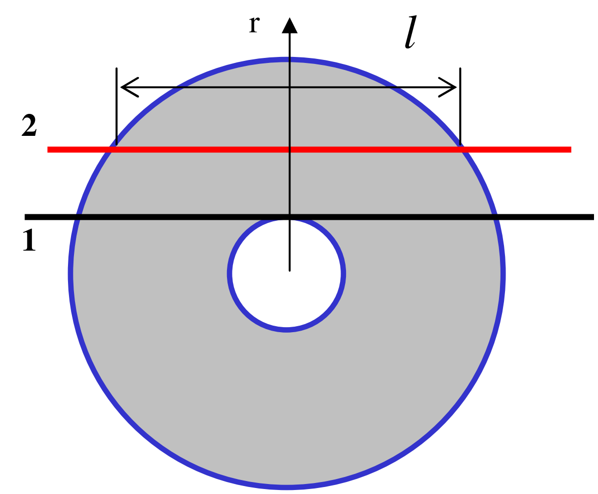
- Location of interferometer laser beams through plasma:

  Midplane: z_2=0; r_2=15 cm

  Off-midplane: z_1=85 cm; r_1=6 cm

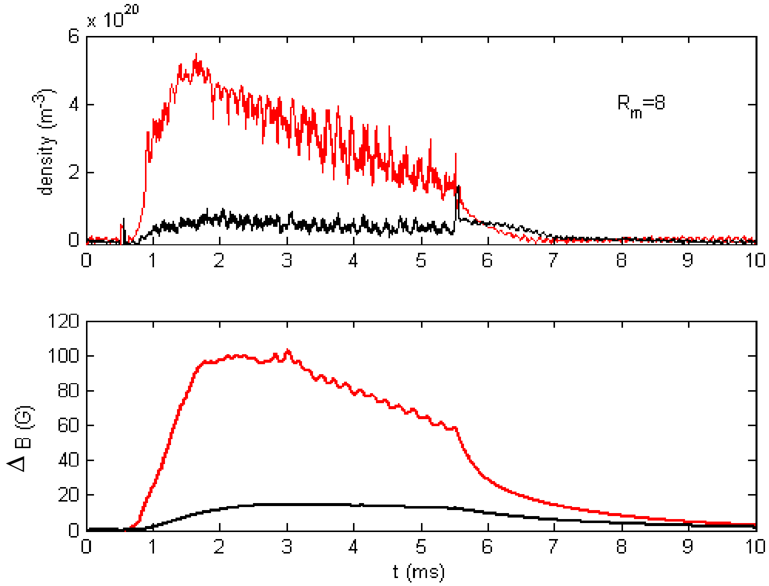




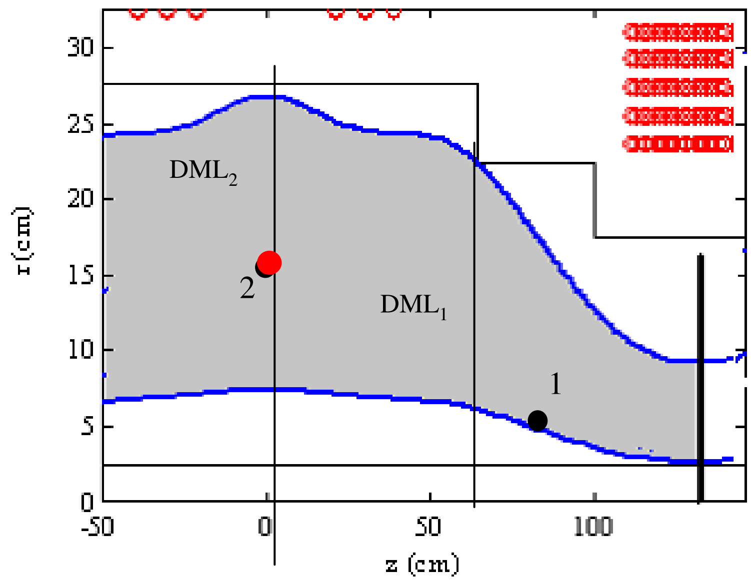


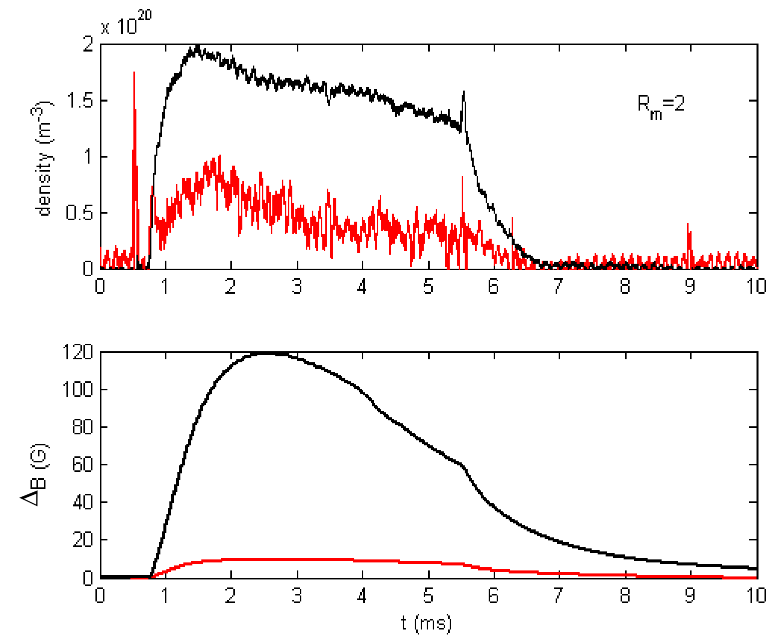
Teodorescu, et al, Phys. Rev. Lett. 105, 085003 (2010)

Plasma density and diamagnetic flux are large at the magnetic minimum

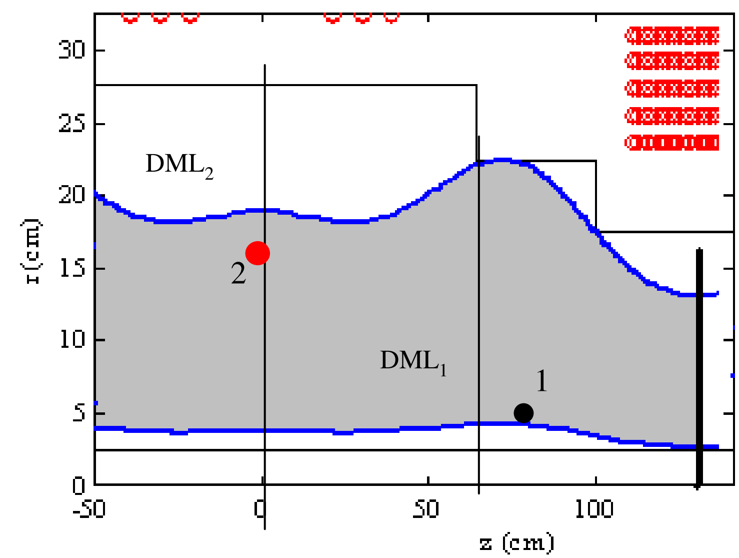


n_2/n_1=12

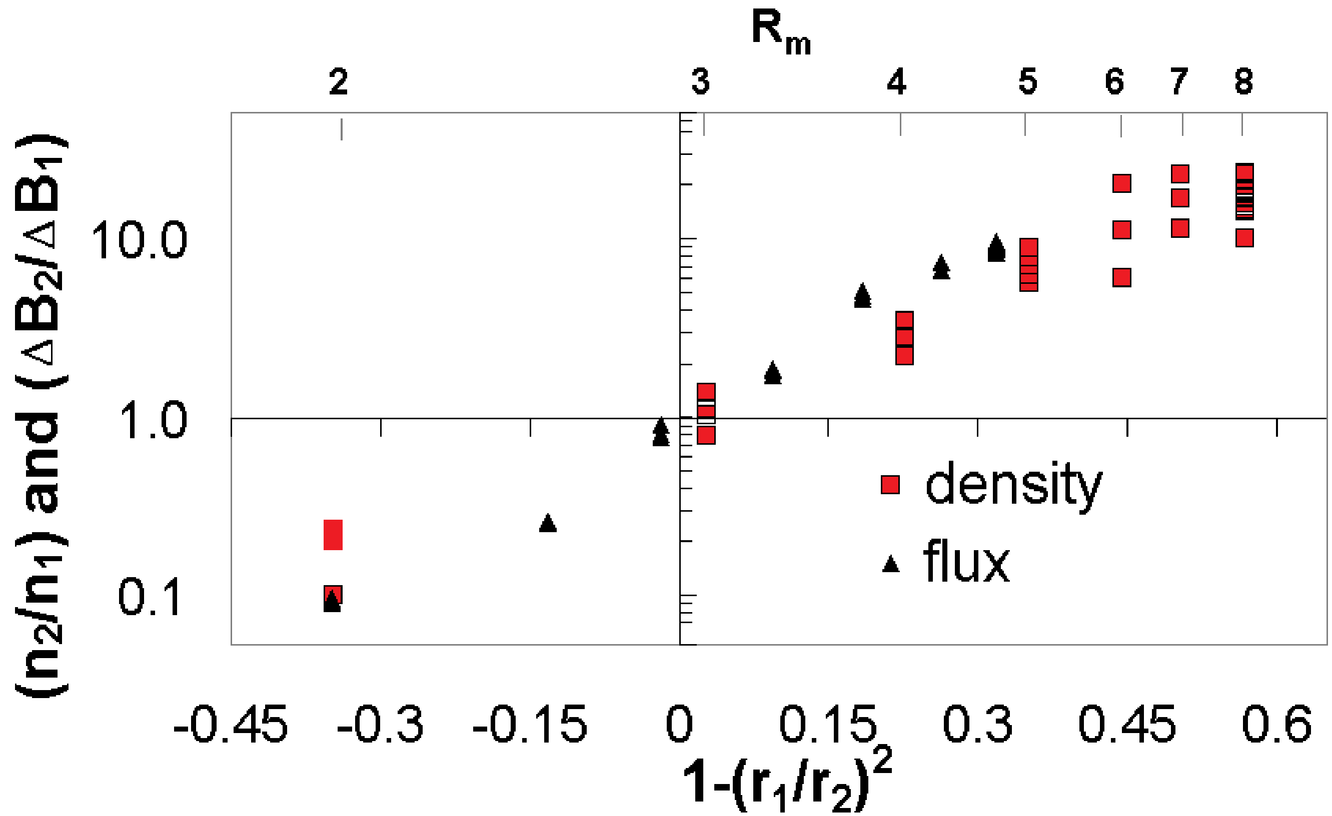


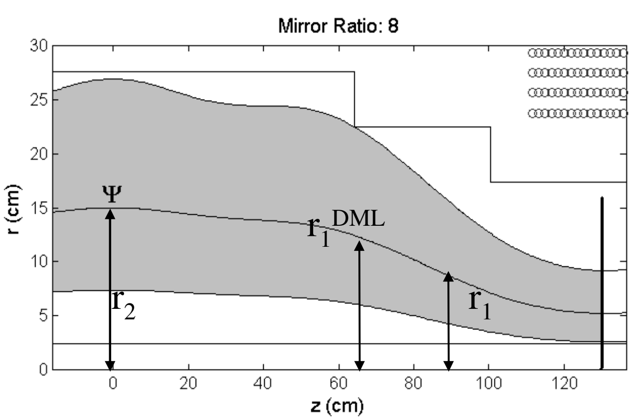


n_2/n_1=0.4

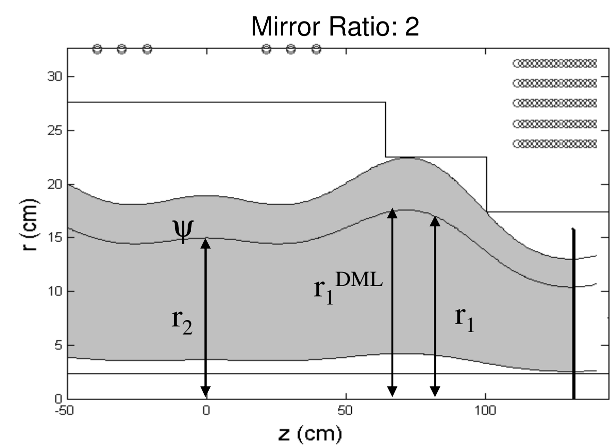


Density ratio and diamagnetic flux ratio flip when $r_1 = r_2$, consistent with radial stratification





• Average values over 100 µs (one momentum confinement time) at *t*=2 ms in the discharge.



0-D Transport Model

$$nMu_{\phi}^{2}/\tau_{mom} = P_{in}$$

$$3nT/\tau_{heat} = P_{in} - P_{rad}$$

$$1/\tau_{mom} = 1/\tau_{perp,i}$$

$$1/\tau_{heat} = 1/\tau_{perp,i} + 1/\tau_{||e}$$

- **0-D Scaling to reactor** ($u_{\phi} < v_A$, classical, Rm=4):

  n=.6 10^{20}, B=2.6T, a=1.1m, P_{in}=3MW

=> T=13keV, M_s=6, P_{fusion}=240MW

UNANSWERED QUESTIONS

- How large, compared to classical, is the residual transport? Is it interchange modes, or other?
- Is there a speed barrier (CIV)? Can it be exceeded?
- $M_s = 6$? M_s not directly controllable
- Insulators at fusion conditions. ~ 10 MV/m?
- Run without core
- Can neutrals be held down?
- Opportunity: High-Tc High-B magnets

A strategic element for fusion

- The system shows
  - ideal MHD equilibrium, with confinement
  - ideal MHD stability
  - steady state, quiescent
  - pathway to Lawson under classical confinement
  - simplicity

- A concept based on small Larmor radius must show, theoretically, at minimum:
  - *Ideal* MHD equilibrium and stability
  - Access to Lawson under *(neo)classical* transport

- The centrifugal concept shows high potential for fusion. This system is underexplored.

Extras

GDT and MCX pressures in same ballpark

GDT: $n \sim 10^{19}\ /\mathrm{m}^3$ $\quad T_e \sim 900$ eV

MCX: $n \sim 4\ 10^{20}\ /\mathrm{m}^3$ $\quad T_i \sim 40$ eV, $T_e \sim 10$ eV

$$nT_{MCX} : nT_{GDT} \sim 40 : 90$$

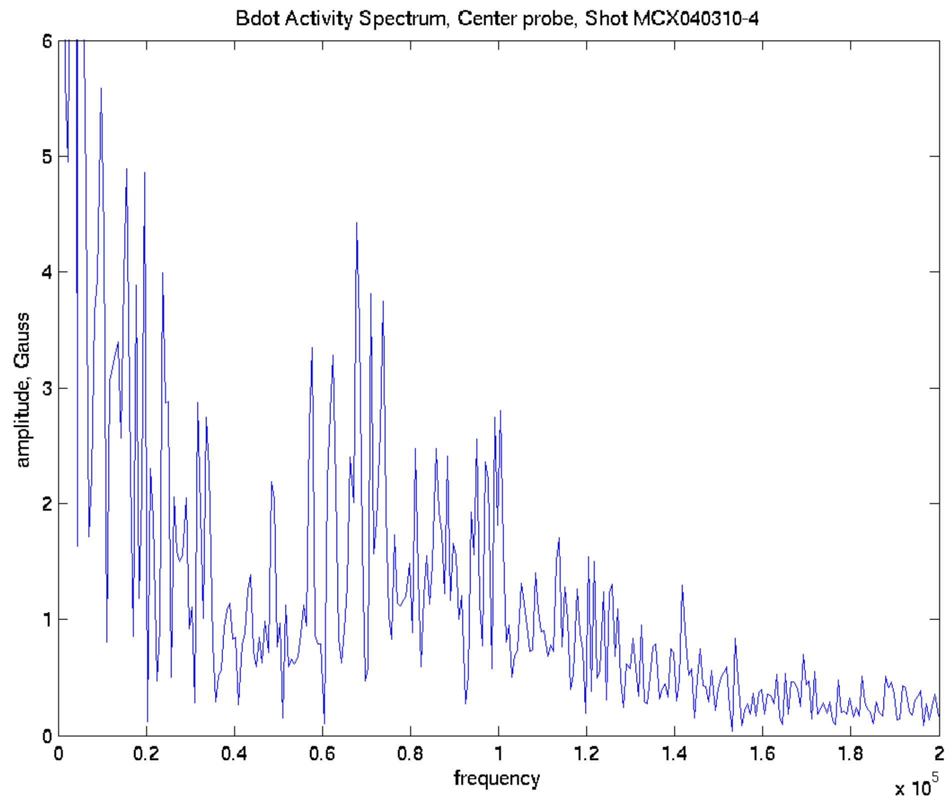
Magnetic probes could yield info on wobbles at the edge

$\delta p \; + \; B\delta B/\mu_0 \; \approx \; 0$

$\delta p \; \sim \; p' \, \delta r$

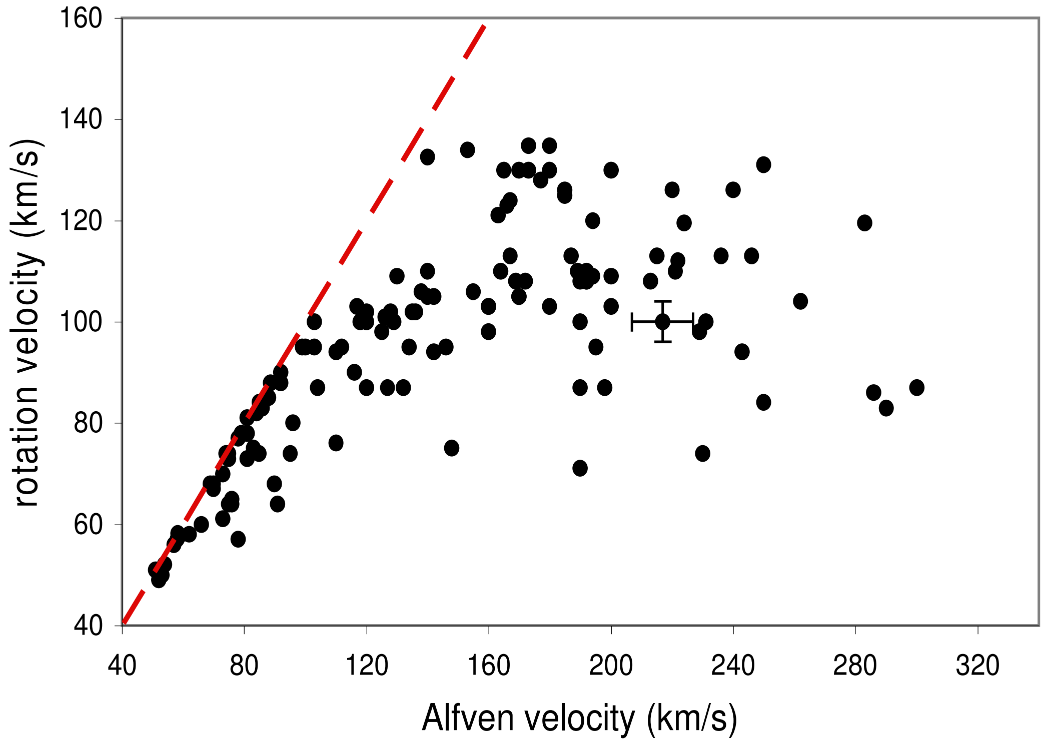
$\delta r/a \; \sim \; B\delta B/\mu_0 p$

$\Rightarrow \delta r < 1$ cm



There is a speed barrier at V_A as expected, but also another non-MHD barrier

$M_A \leq 1$ in all 142 distinct data points



Rotation velocity measured at maximum V_p.

Average values:

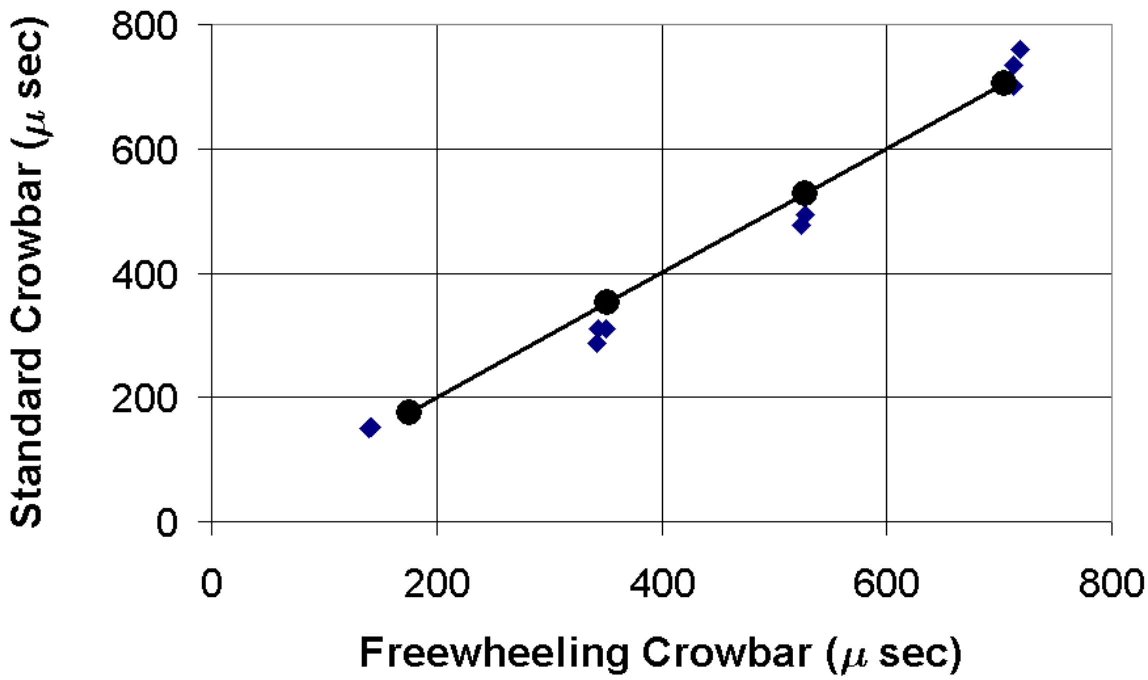
$$u_\phi = \frac{\langle V_p \rangle_\tau}{aB}$$

$$V_A = \frac{B}{(\mu m_i \langle n_i \rangle_\tau)^{1/2}}$$

$$\tau \propto 150 \ \mu s$$

- Consistent with “Critical Ionization Velocity” observed earlier

τ_{momentum} ~ 200-800 μs



=> $N \sim 10^{17}\, m^{-3}$

Calculated timescales for comparison to MCX discharge duration (> 5 ms) and momentum confinement time(200 μs)

($n = 2 \times 10^{20}\ m^{-3}$, $T = 30$ eV, $B = 0.2$ T)

Axial Alfven time ~ L_P/v_A	5μs
Period of rotation ~ $(2\pi R/u_\phi)$	10μs
Interchange growth time ~ $[(a_P L_P)/(T/M_p)]^{1/2}$	**10μs**
Axial electron heat conduction time ~ $(L_P/\lambda)^2\ \tau_e$	30μs
Axial sonic time ~ $L_P/(T/M_p)^{1/2}$	30μs
Electron-ion heat exchange time ~ $(M_p/m_e)\tau_e$	40μs
Classical viscous damping time ~ $(a_P/\rho)^2\ \tau_{ii}$	**8000μs**

Charge exchange time ~ 500 μs

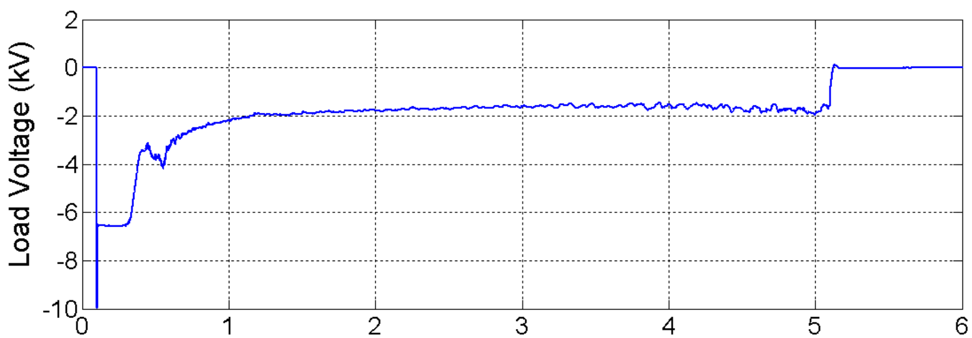
Previous Experiments

- **IXION (LosAlamos) Boyer, et al ‘58**
  - mirror geometry
  - ExB rotation as expected ~ 40 km/s
  - impurity influx terminates discharge

- **F-X (Stockholm) Lehnert, et al ‘60’s**
  - dipole geometry
  - plasma dielectric as expected
  - $V_0 < 10$ keV limitation, arcing @ insulator

- **PSP (Novosibiirsk) Volosov, et al ‘70’s**
  - biased, concentric ring electrodes => high V_0
  - line-tied stabilization
  - high T, $n \sim 2\ 10^{17}$ /m^3

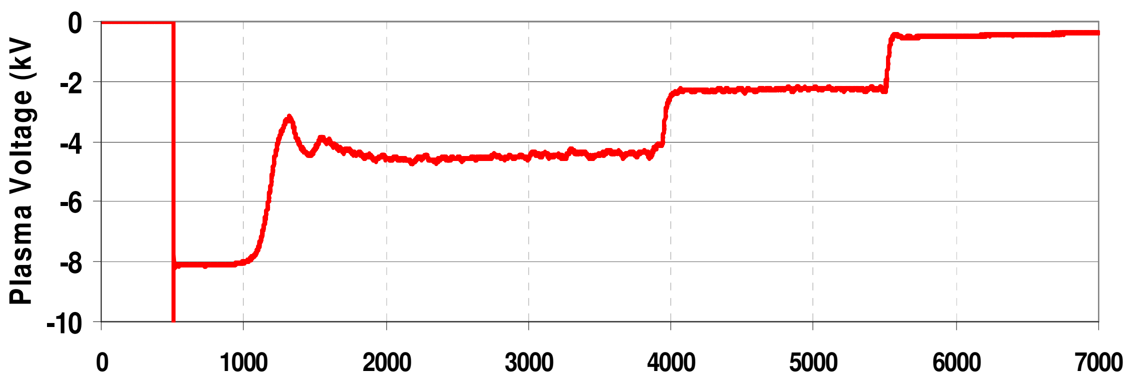
HR -mode discovered



Rotation speed H ≥ 3

Mach number H ~ 3

Confinement time H ~ 3



References (partial list)

1) Sub-Alfvenic velocity limits in magnetohydrodynamic rotating plasmas. Physics of Plasmas 17 052503 (2010)

C. Teodorescu, R. Clary, R. F. Ellis, A.B. Hassam, C. A. Romero-Talamas and W.C. Young.

2) Low Dimensional Model for the Fluctuations observed in the Maryland Centrifugal Excperiment. International Symposium of Waves, Coherent Structures and Turbulence in Plasmas, 2010 American Institute of Physics 978-0-7354-0865-4/10

P.N.Guzdar, I. Uzun-Kaymak, A.B.Hassam, C. Teodorescu, R.F. Ellis, R.Clary, C.Romero-Talamas, and W. Young

3) Isorotation and differential rotation in a magnetic mirror with imposed ExB rotation. Physics of Plasmas 19, 072501 (2012). C.A. Romero-Talamas, R.C. Elton, W.C. Young, R. Reid and R.F. Ellis.

4) Experimental study on the velocity limits of magnetized rotating plasmas. Physics of Plasmas 15 042504 (2008). C. Teodorescu, R. Clary, R.F. Ellis, A.B. Hassam, R. Lunsford

5) Diamagnetism of rotating plasma. W.C. Young, A.B. Hassam, C.A. Romero-Talamas, R.F.Ellis and C. Teodorescu.

Physics of Plasmas 18, 112505 (2011)

6) Analysis and modeling of edge fluctuations and transport mechanism in the Maryland Centrifugal Experiment. I.U.Uzun-Kaymak, P.N. Guzdar, R. Clary, R.F.Ellis, A.B. Hassam and C. Teodorescu. Physics of Plasmas 15, 112308 (2008)

7) 100 eV electron temperatures in the Maryland centrifugal experiment observed using electron Bernstein emission. R.R. Reid, C.A. Romero-Talamas, W.C.Young, R.F.Ellis, and A.B.Hassam. Physics of Plasmas 21, 063305 (2014)

8) Confinement of Plasma along Shaped Open Magnetic Fields from the Centrifugal Force of Supersonic Plasma Rotation. C. Teodorescu, W.C.Young, G.W. Swan, R.F.Ellis, A.B.Hassam, and C.A.ROmero-Talamas. Phys. Rev. Lett. 105, 085003 (2010)

9) Charge and mass considerations for plasma velocity measurements in rotating plasmas. C.A. Romero-Talamas, R.C.Elton, W.C. Young, R. Reid, R.F.Ellis, A.B. Hassam. Journal of Fusion Energy, 29, 6, 543-547 (2010)